

Introduction to Microeconomics

Consumer Theory: Preferences and Budgets

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The Law of Demand

People do less of what they want to do as the cost of doing it rises.

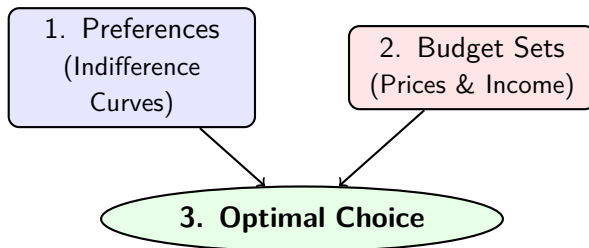
 The cost of an activity, good, or service involves not just monetary costs, but nonmonetary costs as well.

- The goal is to provide a theory of consumer choice based on some reasonable assumptions about the consumer.
- What we will study is part of the workhorse models in Microeconomics and also one of the foundations of modern Macroeconomic Theory.
- Our approach will be to study the consumer embedded in a market Economy, where there are prices for goods and services, as well as wealth constraints.
- However, at its core, the approach is quite general and can be used to analyze a wide variety of problems, not only limited to what is traditionally understood as Economics, but from other social sciences as well.

The Three Ingredients of Consumer Choice

To understand how a consumer behaves, we combine three distinct components:

- ❶ **Consumer Preferences:** The "What I want" component. It describes how a consumer ranks different bundles of goods based on their tastes.
- ❷ **Budget Sets:** The "What I can afford" component. This is determined by the consumer's income and the prices of goods in the market.
- ❸ **The Optimization:** The "Decision." We assume the consumer chooses the *best* bundle they can actually afford.



***Pedagogical Note:** While we study these separately, "The Law of Demand" emerges only when we change the param-*

A model of consumer Preferences

- We start with a set of objects (bundles) X . Elements from from this set, like x or y , are the choices that the consumer can make.
- Most applications consider $X \subset \mathbb{R}_+^n$.
- In this class, $X = \mathbb{R}_+^2$. A bundle x is then $x = (x_1, x_2)$. Interpret it as: x_1 is the quantity of good 1 and x_2 is the quantity of good 2.

Consumer preferences: The Questionnaire Q

Consists of a long list of questions of the following form.

$Q(x, y)$, x and y are different bundles. How do you compare x to y ? Check one and only one of the following three options:

- ☐ I strictly prefer x to y ($x \succ y$).
- ☐ I strictly prefer y to x ($y \succ x$).
- ☐ I am indifferent between x and y ($x \sim y$).

A model of consumer preferences is just a long list of questions of this type and their answers!

What is a legal answer to the questionnaire?

- We need to put some restrictions (assumptions) on answers.

Restriction 1: Completeness

For any question $Q(x, y)$ exactly one of the boxes is checked. We don't allow

- Refraining to answer (no box checked).
- Checking more than one box.
- This means we exclude attitudes like:
 - They are not comparable.
 - I don't know what x is.
 - I have no opinion.
 - I prefer x over y and y over x .
- We are also not allowing attitudes like:
 - It depends on what others think.
 - It depends on the circumstances (frame).
 - Any consideration that includes **intensity**, like I somewhat prefer x , or I love x and hate y .

Restriction 2: Transitivity

Suppose I have bundles x, y, z such that $Q(x, y) : x \succ y$, and $Q(y, z) : y \succ z$. Then this implies that your answer to $Q(x, z)$ should be $x \succ z$.

Similarly, if $Q(x, y)$ is $x \sim y$ and $Q(y, z)$ is $y \sim z$ then in $Q(x, z)$ your answer should be $x \sim z$.

The "At Least As Good" Relation

- Questionnaire Q can be rephrased using the **weak preference** relation:

- $Q(x, y)$: How do you compare x to y ? Check at least one:
 - ☐ x is *at least as good as* y ($x \succsim y$).
 - ☐ y is *at least as good as* x ($y \succsim x$).

- Consistency check:**

- If $x \succsim y$ and $y \succsim x$, then $x \sim y$ (Indifference).
- If $x \succsim y$ but $y \not\succsim x$, then $x \succ y$ (Strict Preference).

- The Axioms of Preferences (Rationality):**

Completeness: For any two bundles $x, y \in X$, either $x \succsim y$, $y \succsim x$, or both.

Transitivity: For any three bundles $x, y, z \in X$, if $x \succsim y$ and $y \succsim z$, then it must be that $x \succsim z$.

The "Money Pump"

If transitivity fails (e.g., $x \succ y$, $y \succ z$, but $z \succ x$), a clever merchant could trade you y for x plus a penny, then z for y plus a penny, then x for z plus a penny... cycling until you are broke!

Preference Relations

Any relation R over X is called a **preference relation** if it consists of a set of legal answers to the questionnaire Q , that is, if it satisfies completeness and transitivity.

Other legal answers to Q : Restrictions

Suppose the set of bundles $X \subset \mathbb{R}_+^n$ consists of *goods that are desirable*.

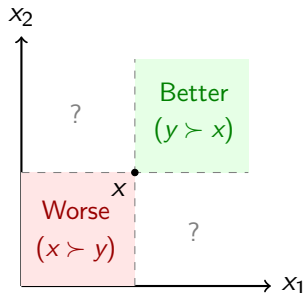
Restriction 3: Monotonicity or “more-is-better”

If $x \geq y$, where $\geq$ denotes the component-wise standard order, then $x \succsim y$.

If $x > y$, where $>$ denotes the component-wise strict order, then $x \succ y$.

Example: say good 1 is food and good 2 is clothing. $x = (30, 40)$ and $y = (20, 30)$. If preferences satisfy monotonicity, then the answer to $Q(x, y)$ should be $x \succ y$.

- **Preferred Set:** All bundles to the North-East of x are strictly better.
- **Dominated Set:** All bundles to the South-West of x are strictly worse.
- **The Trade-off:** For a bundle z to be indifferent to x ($z \sim x$), it *must* lie in the North-West or South-East quadrants.



Indifference curves

- For a bundle x the **indifference curve at x** is the set of all bundles that are indifferent to x :

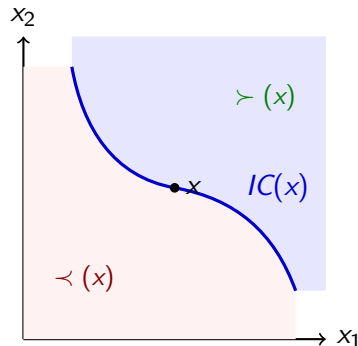
$$IC(x) = \{y \in X : y \sim x\}$$

- The strictly preferred set to x :

$$\succ(x) = \{y \in X : y \succ x\}$$

- The strictly worse set to x :

$$\prec(x) = \{y \in X : x \succ y\}$$

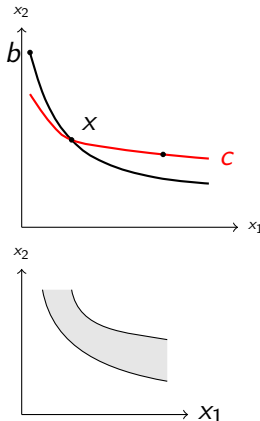


Insight: What happens if you add monotonicity?

Notice that $IC(x)$ is the "frontier." Under monotonicity, moving North-East always puts you in $\succ(x)$, while moving South-West always puts you in $\prec(x)$.

Implications of the restrictions

- Completeness $\Rightarrow$ map is **full**.
- Transitivity $\Rightarrow$ **indifference curves do not intersect**.
- Monotonicity $\Rightarrow$ **indifference curves are not thick**.



Transitivity Violation: If $b \sim x$ and $c \sim x$. How are b and c compared?

- if $b \succ c$, then $c \sim x$ and $x \sim b \Rightarrow b \sim c$, contradiction!
- if $c \succ b$, then $c \sim x$ and $x \sim b \Rightarrow c \sim b$, contradiction!
- If $b \sim c$, then $c \sim x$ and $x \sim b \Rightarrow b \sim c$, but they are on different IC's, contradiction!

Monotonicity Violation: In a "thick" IC, you can find two bundles where one has strictly more of both goods but is ranked the same.

From Preferences to Utility

- Although preference relations provide a model of the consumers preferences, they are somewhat cumbersome to work with.
- Recall: a preference relation provides an **ordering** over the consumption bundles.
- Idea: can we get some mathematical object that is more tractable to work with, while still preserving the idea that a preference relation provides an ordering over consumption bundles?
- The answer is **Yes**!
- We say that a **utility function** $u : X \rightarrow \mathbb{R}$ represents the preference relation $\succsim$ over X if for every $x, y \in X$, $x \succsim y \Leftrightarrow u(x) \geq u(y)$.
- Given a preference relation, can we always ensure that a utility function exists?
 - If $\succsim$ is complete and transitive and X is finite or (countable), the answer is Yes.
 - On the other hand, if X is infinite, like $\mathbb{R}_+^2$, then we need an extra technical restriction: **continuity**, but this will be discussed in higher level courses.

Utility is an ordinal concept, not a cardinal one.

- Suppose that a preference relation $\succsim$ can be represented by the utility function u .
- Do the numbers $u(x)$ for each $x \in X$ matter?
- The answer is no, what matters is the order (Ranking), not the number!
- Utility is ordinal, not cardinal (the magnitude matters).
- The previous idea can be formally stated with the concept of a **monotonic transformation**:
 - Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be an increasing function, i.e. if $a \geq b$ then $f(a) \geq f(b)$.
 - Let u be a utility function representing $\succsim$. Then $v(x) = f(u(x))$ represents the same preferences as u .
 - $x \succsim y \Leftrightarrow u(x) \geq u(y) \Leftrightarrow v(x) \geq v(y)$.

Monotonic Transformations

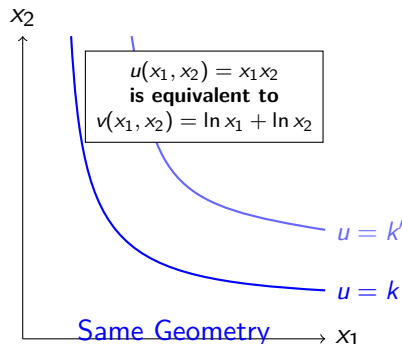
- A **monotonic transformation** is a function $f(u)$ that is strictly increasing.
- If $f(u)$ is a monotonic transformation, then $v = f(u(x_1, x_2))$ represents the **same preferences** as $u(x_1, x_2)$.
- The ranking of bundles remains unchanged!

The Log Transformation

If $u(x_1, x_2) = x_1 x_2$, then:

$$v = \ln(u) = \ln(x_1) + \ln(x_2)$$

represents the same indifference curves.



From Utility to Indifference curves

- With the help of a utility function, it becomes easy to define an indifference curve:

$$x \sim y \Leftrightarrow u(x) = u(y)$$

- Then the indifference curve passing through x is the set of consumption bundles that achieve the same level of utility $u(x)$:

$$IC(x) = \{y \in X : u(y) = u(x)\}$$

- We can define in a similar way the strictly preferred and the strictly worse sets:

- **Strictly Preferred Set:**

$$\succ(x) = \{y \in X : u(y) > u(x)\}$$

- **Strictly Worse Set:**

$$\prec(x) = \{y \in X : u(y) < u(x)\}$$

Note on Level Sets

In mathematical terms, an indifference curve is simply a **level set** of the utility function, i.e., $u(x) = k$, where $x = (x_1, \dots, x_n)$ and k is a real number.

Example: From Utility to Indifference Curves

Consider the utility function:

$$u(x_1, x_2) = x_1 x_2$$

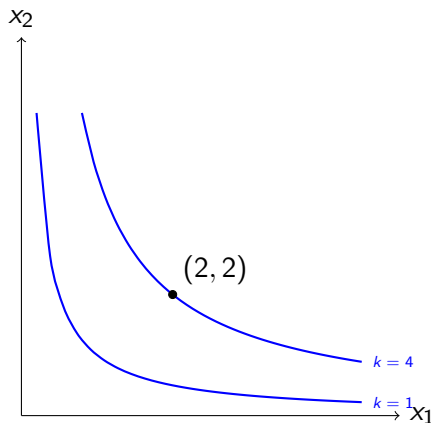
- To find the indifference curve for a utility level k , we set:

$$u(x_1, x_2) = k \implies x_1 x_2 = k$$

- Solving for x_2 gives us the function for the IC:

$$x_2 = \frac{k}{x_1}$$

- For $k = 1$, the IC is $x_2 = 1/x_1$.
- For $k = 2$, the IC is $x_2 = 2/x_1$.
- Higher values of k result in curves further from the origin.



The Marginal Rate of Substitution

- It is the absolute value of the slope of the indifference curve at a particular point.
- It measures the cost of opportunity of good 1 in terms of good 2 while staying on the same indifference curve.
- Intuitively, it is the personal opportunity cost (in terms of the consumer's preferences) of remaining in the same indifference curve, measured as units of good 2 that must be given up to get one more unit of good 1.
- How? Let's compute the total change in utility, when the utility function is $u(x_1, x_2)$:

$$\Delta U = \frac{\Delta U}{\Delta x_1} \Delta x_1 + \frac{\Delta U}{\Delta x_2} \Delta x_2$$

- At an indifference curve, utility is not changing, so $\Delta U = 0$, this leads to:

$$\frac{\Delta x_2}{\Delta x_1} = - \frac{\frac{\Delta U}{\Delta x_1}}{\frac{\Delta U}{\Delta x_2}}$$

- The **Marginal Rate of Substitution** is the absolute value of the quantity above.

The Marginal Rate of Substitution: formal derivation.

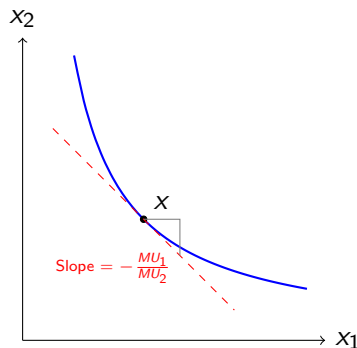
- The **MRS** is the absolute value of the slope of the indifference curve.
- By the **Implicit Function Theorem**, an IC defined by $u(x_1, x_2) = k$ implicitly defines x_2 as a function of x_1 .
- Differentiating both sides with respect to x_1 :

$$\frac{\partial u}{\partial x_1} \frac{dx_1}{dx_1} + \frac{\partial u}{\partial x_2} \frac{dx_2}{dx_1} = 0$$

- Which simplifies to:

$$MU_1 + MU_2 \frac{dx_2}{dx_1} = 0$$

- Solving for the slope of the IC: $\frac{dx_2}{dx_1} = -\frac{MU_1}{MU_2}$
- The MRS is the absolute value: $MRS_{1,2} = \frac{MU_1}{MU_2}$



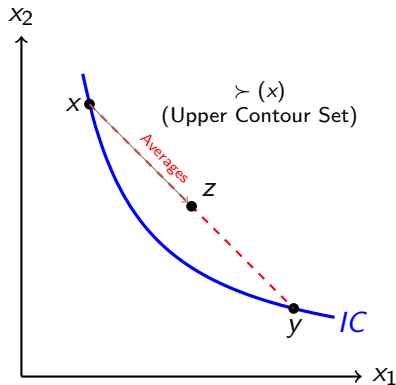
Convexity of Preferences

- **Intuition:** Consumers prefer "mixtures" or averages over extreme bundles.
- **Formal Definition:** A preference relation is *convex* if for any $y \sim x$, the bundle:

$$z = tx + (1 - t)y$$

is at least as good as x for all $t \in [0, 1]$.

- **Strict Convexity:** If $x \neq y$ and $y \sim x$, then $z \succ x$.
- Geometrically, the **Upper Contour Set** is a convex set.



The budget set and the budget line for the consumer.

- Goal: represent what is feasible to consume given market prices and income.
- Consumption space: $X = \mathbb{R}_+^n$, n goods, indexed x_i , $i = 1, \dots, n$.
- Prices $p_1, \dots, p_n$ for each good, all non-negative.
- Consumer takes prices as given, i.e., consumer cannot “bargain” for a different price.
- The budget set is the set of all **feasible** consumption bundles given prices and income I of the consumer:

$$B(p_1, \dots, p_n, I) = \{(x_1, \dots, x_n) : p_1x_1 + p_2x_1 + \dots p_nx_n \leq I\}$$

- Intuitively: Total expenditures must be less than or equal to Income.
- When $p_1x_1 + \dots + p_nx_n = I$, this corresponds to the **budget hyperplane**, and is the set of consumption bundles that are feasible and exhaust the consumer's income.

The two-good case

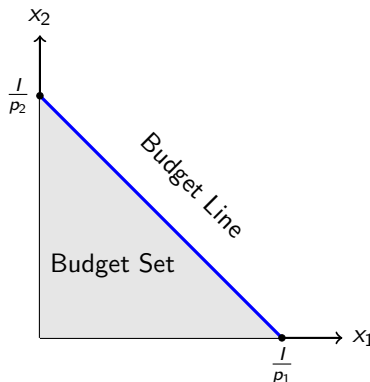
- Suppose we are only considering two goods, i.e., $X = \mathbb{R}_+^2$, goods x_1 and x_2 .
- Prices are p_1 and p_2 and income I .
- The budget set is described by the simple inequality: $p_1x_1 + p_2x_2 \leq I$.
- The budget line is $p_1x_1 + p_2x_2 = I$ is the set of feasible consumption bundles which exhaust the consumer's Income.

Slope Derivation:

$$p_2x_2 = I - p_1x_1$$

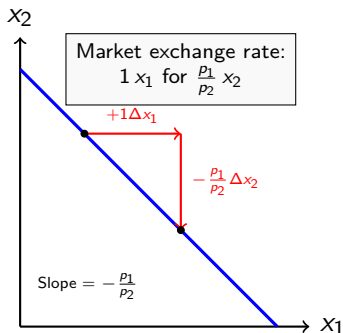
$$x_2 = \frac{I}{p_2} - \frac{p_1}{p_2}x_1$$

The slope is $-\frac{p_1}{p_2}$.



Economic Interpretation of the slope

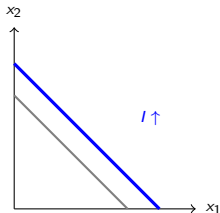
- The slope of the budget line ($-\frac{p_1}{p_2}$) represents the **market opportunity cost** of good 1 in terms of good 2.
- Even if goods are quoted in money, the **relative price ratio** tells us the trade-off:
 - To buy 1 more unit of x_1 , you must spend p_1 dollars.
 - Those p_1 dollars could have bought $\frac{p_1}{p_2}$ units of x_2 .
- **Connection to Trade:** This is exactly the same logic as *trading when there is comparative advantage*.
- The market acts as a "trading partner" that offers to swap goods at a fixed rate of exchange: 1 unit of x_1 for $\frac{p_1}{p_2}$ units of x_2 .



Changes in the Budget Line

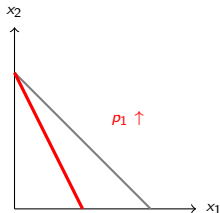
1. Changes in Income (I)

- If I increases $\implies$ **Parallel shift** outward.
- If I decreases $\implies$ **Parallel shift** inward.
- **Why?** The slope ($-p_1/p_2$) does not change.



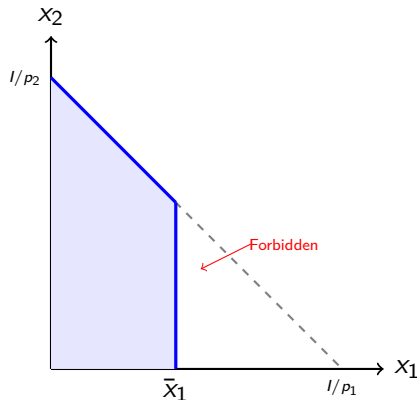
2. Changes in Prices (p_1)

- If p_1 increases $\implies$ **Pivot** inward.
- If p_1 decreases $\implies$ **Pivot** outward.
- **Why?** The slope ($-p_1/p_2$) and one intercept change.



Non-linear Budget Constraints: Quantity Restrictions

- In some markets, the consumer faces a limit on the maximum amount of a good they can purchase.
- Let $\bar{x}_1$ be the **rationing level** for good 1.
- The budget set is now defined by two simultaneous constraints:
 - ① $p_1x_1 + p_2x_2 \leq I$
 - ② $x_1 \leq \bar{x}_1$
- Bundles where $x_1 > \bar{x}_1$ are **non-feasible**, even if the consumer can afford them.



Budget Constraint with Quantity Discounts

Pricing of good x_1 :

- $p_1 = p_1^H$ if $x_1 \leq \bar{x}$
- $p_1 = p_1^L$ if $x_1 > \bar{x}$ ($p_1^L < p_1^H$)

The Budget Line:

- If $x_1 \leq \bar{x}$:

$$x_2 = \frac{I}{p_2} - \frac{p_1^H}{p_2} x_1$$

- If $x_1 > \bar{x}$:

$$p_1^H \bar{x} + p_1^L (x_1 - \bar{x}) + p_2 x_2 = I$$

Solving for x_2 :

$$x_2 = \frac{I - (p_1^H - p_1^L)\bar{x}}{p_2} - \frac{p_1^L}{p_2} x_1$$

- **Kink:** Occurs at $x_1 = \bar{x}$.
- **Slope:** Flattens from $-\frac{p_1^H}{p_2}$ to $-\frac{p_1^L}{p_2}$.

